DATA COMMUNICATION AND COMPUTER NETWORKS

T1. Behrouz A. Forouzan, “Data Communications and Networking with TCP/IP Protocol Suite”, 6th Edition, Tata McGraw-Hill, 2021

DATA COMMUNICATION AND COMPUTER NETWORKS
CSE 2251

Module-1.1

Data Communication and Network Model

DATA COMMUNICATION

- **Data communications and networking are changing today's world. Why wait for a week for some report from UK to arrive by post where you can receive instantaneously through computer networks?**

- **The term telecommunication means communication at a distance.**

- **The word data refers to information presented in whatever form is agreed upon by the parties creating and using the data.**

- **Data communications are the exchange of data between two devices via some form of transmission medium such as a wire cable.**

CHARACTERISTICS OF DC

- ***The four fundamental characteristics of data communications are:-***

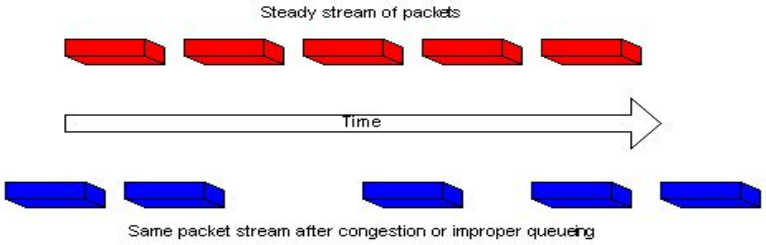
- **Delivery**: The system must deliver data to the correct destination. Data must be received by the intended device or user and ***only*** by that device or user.

Accuracy: The system must deliver the data accurately. Data that have been altered in transmission and left uncorrected are unusable.

CHARACTERISTICS OF DC

- **Timeliness**: The system must deliver data in a **timely manner**. Data delivered late are useless. In the case of video and audio, timely delivery means delivering data as they are produced, in the same order that they are produced, and without significant delay.

Jitter: Jitter refers to the **variation** in the packet arrival time. It is the uneven delay in the delivery of audio or video packets. For example, let us assume that video packets are sent every 30ms. If some of the packets arrive with 30-ms delay and others with 40-ms delay, an uneven quality in the video is the result.



COMPONENTS OF DC

Components

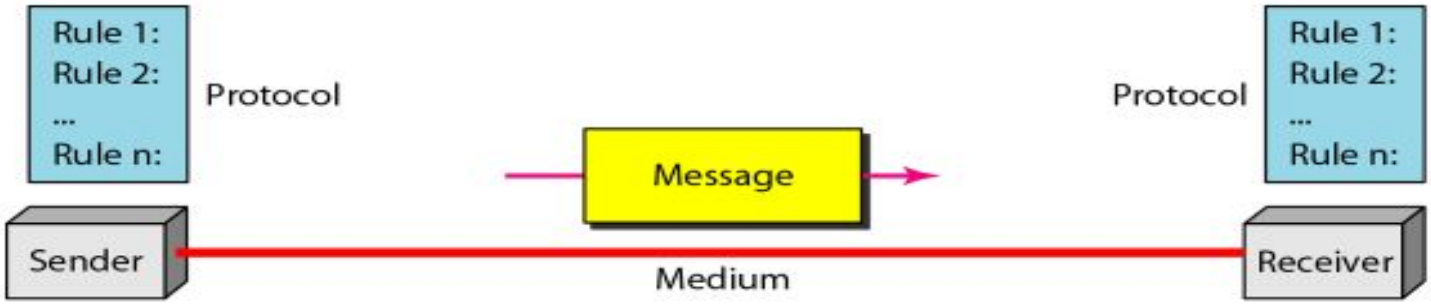


Figure: *Five components of data communication*

COMPONENTS OF DC

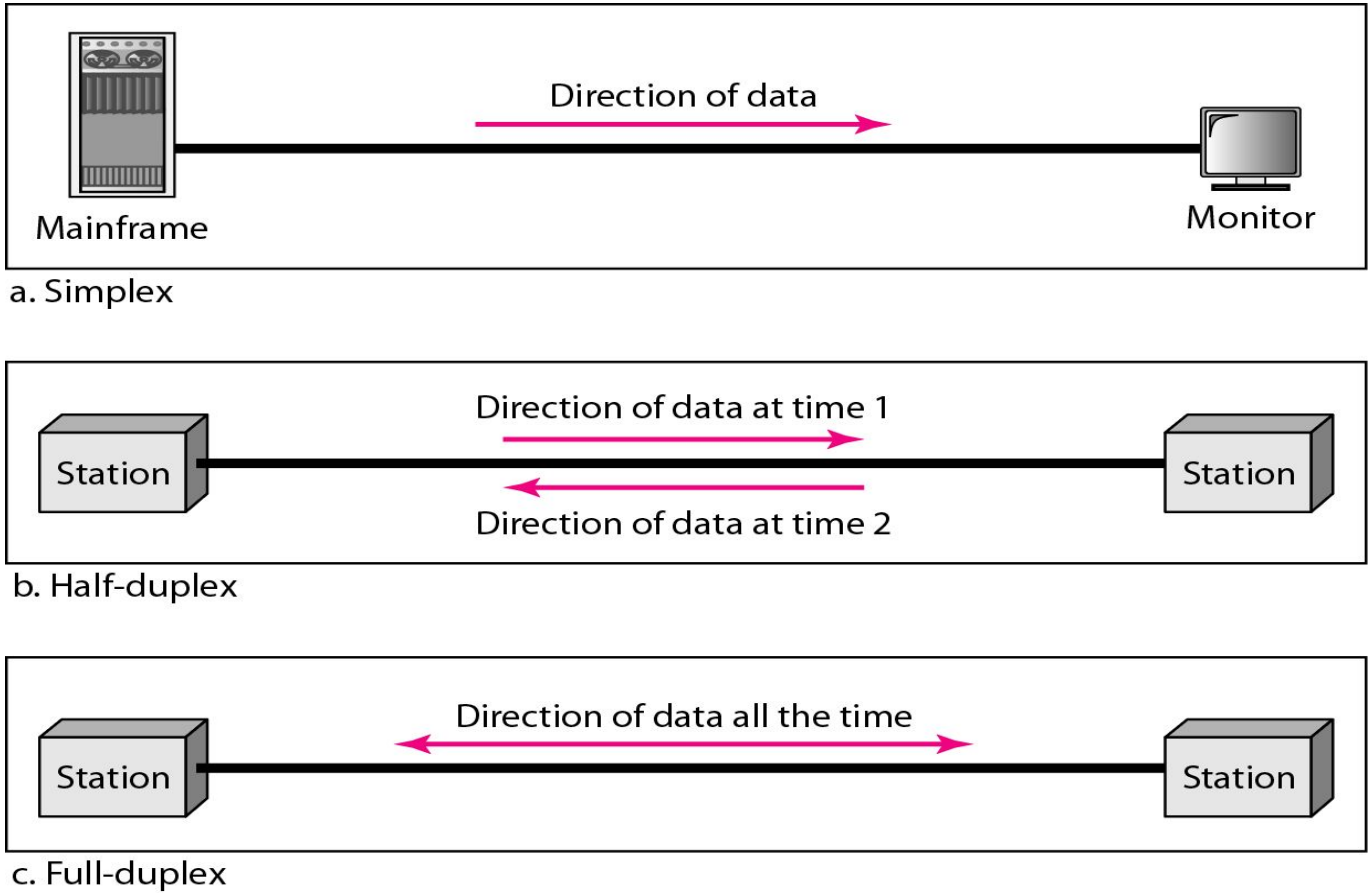
- ***The five components of data communications are:***
- **Message**: The message is the information (data) to be communicated. Popular forms of information include text, numbers, pictures, audio, and video.
- **Sender**: The sender is the device that sends the data message. It can be a computer, workstation, telephone handset, mobile, and so on.
- **Receiver**: The receiver is the device that receives the message. It can be a computer, workstation, telephone handset, television, and so on.
- **Transmission medium**: The transmission medium is the physical path by which a message travels from sender to receiver. Some examples of transmission media include twisted-pair wire, coaxial cable, fibre-optic cable, and radio waves.
- **Protocol**: A protocol is a set of rules that govern data communications. It represents an agreement between the communicating devices. Without a protocol, two devices may be connected but not communicating.

DATA REPRESENTATION

- **Information today comes in different forms such as :**

- Text
- Numbers
- Images
- Audio
- Video

DATA FLOW (SIMPLEX, HALF-DUPLEX, AND FULL-DUPLEX)



DATA FLOW(CNTD)

- Communication between two devices can be **simplex**, **half-duplex**, or **full-duplex** as shown in figure above.

- **Simplex**

- In simplex mode, the communication is **unidirectional**, as on a one-way street.Only one of the two devices on a link can transmit; the other can only receive.
- Example:Keyboards and traditional monitors

- **Half-Duplex**

- In half-duplex mode, each station can both transmit and receive, **but not at the same time**. When one device is sending, the other can only receive, and vice versa.
- In a half-duplex transmission, the entire capacity of a channel is taken over by whichever of the two devices is transmitting at the time.

- **Example:** Walkie-Talkie

DATA FLOW(CNTD)

Full-Duplex

- In full-duplex mode (also called duplex), both stations can transmit and receive **simultaneously**.
- In full-duplex mode, signals going in one direction share the capacity of the link with signals going in the other direction.

Example: Telephone

INTRODUCTION TO NETWORKS

A network is a set of devices (often referred to as nodes) connected by communication links.

A node can be a computer, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network.

A link can be a cable, air, optical fiber, or any medium which can transport a signal carrying information.

NETWORK CRITERIA

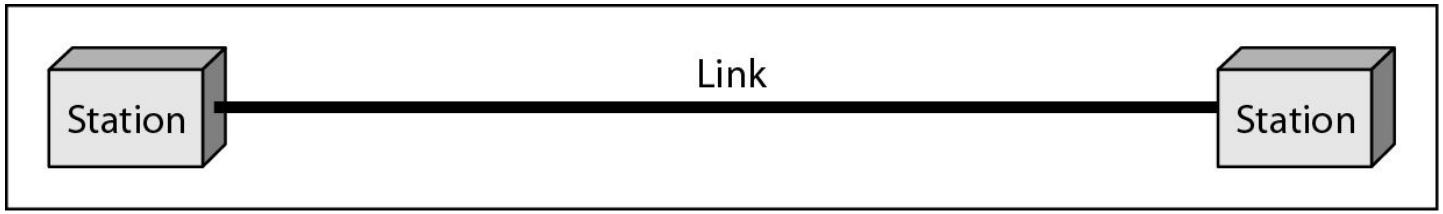
- **Performance**
  - Depends on Network Elements
  - Measured in terms of Delay and Throughput
- **Reliability**
  - Failure rate of network components
  - Measured in terms of availability/robustness
- **Security**
  - Data protection against corruption/loss of data due to:
    - Errors
    - Malicious users

PHYSICAL STRUCTURES

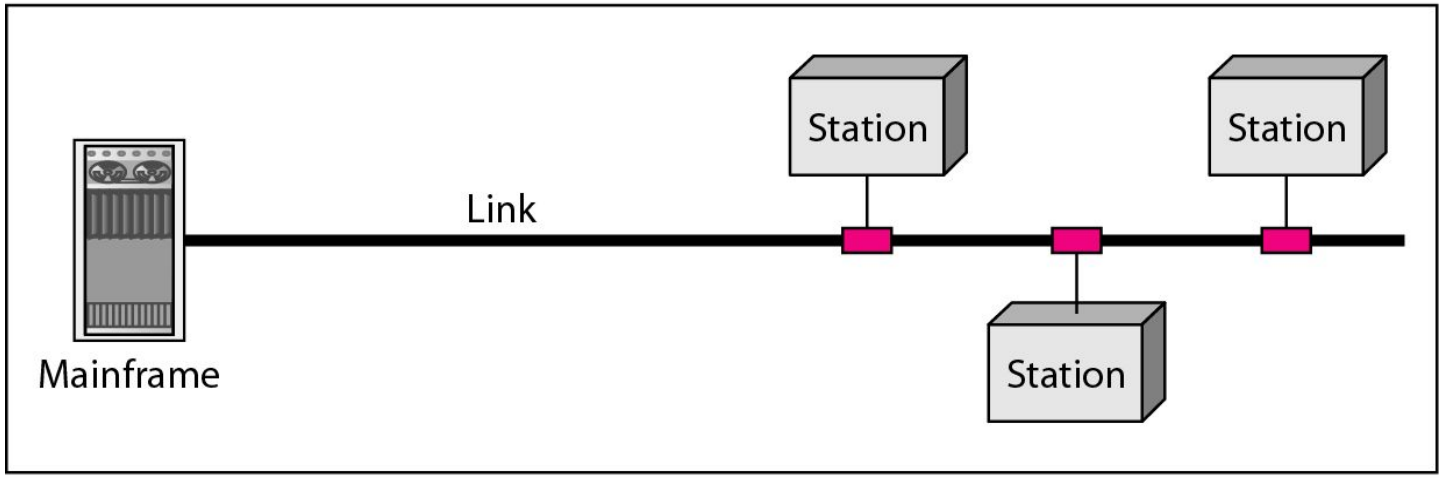
- **Type of Connection**
  - Point to Point - single transmitter and receiver
  - Multipoint - multiple recipients of single transmission

- **Physical Topology**
  - Connection of devices
  - Type of transmission - unicast, multicast, broadcast

Types of connections: point-to-point and multipoint



a. Point-to-point

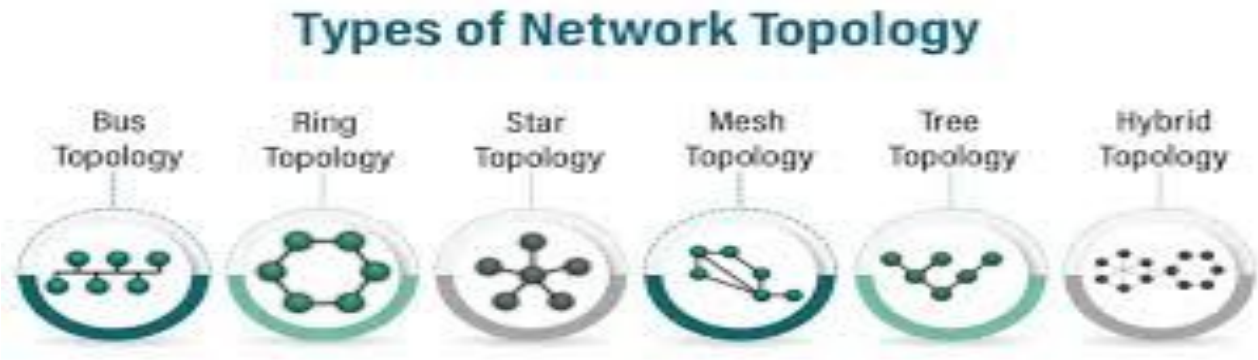


b. Multipoint

NETWORK TOPOLOGY

Network Topology is the schematic description of a network arrangement, connecting various nodes(sender and receiver) through lines of connection.

TYPES OF NETWORK TOPOLOGIES



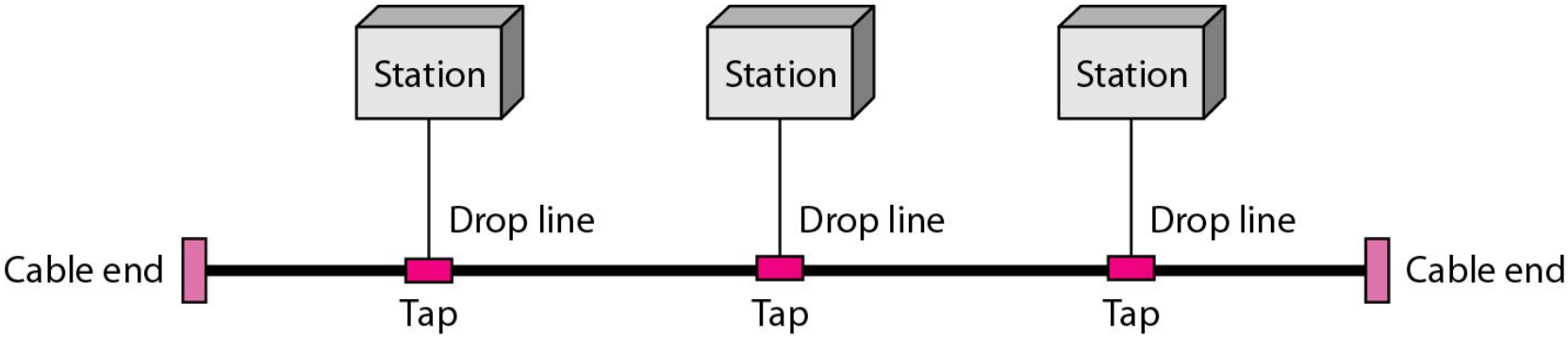
1. **Bus Topology**
2. **Ring Topology**
3. **Star Topology**
4. **Mesh Topology**
5. **Tree Topology**
6. **Hybrid Topology**

BUS TOPOLOGY

Bus topology is a network type in which every computer and network device connected to single cable. When it has exactly two endpoints, then it is called **Linear Bus topology**.

Features of Bus Topology

- It transmits data only in one direction.
- Every device is connected to a single cable



Bus Topology

Advantages of Bus Topology

- It is cost effective and used in small networks.
- It is easy to understand.
- Easy to expand joining two cables together.
- Cable required is least compared to other network topology.

Disadvantages of Bus Topology

- Cables fails then whole network fails.
- If network traffic is heavy or nodes are more the performance of the network decreases.
- Cable has a limited length.
- It is slower than the ring topology.

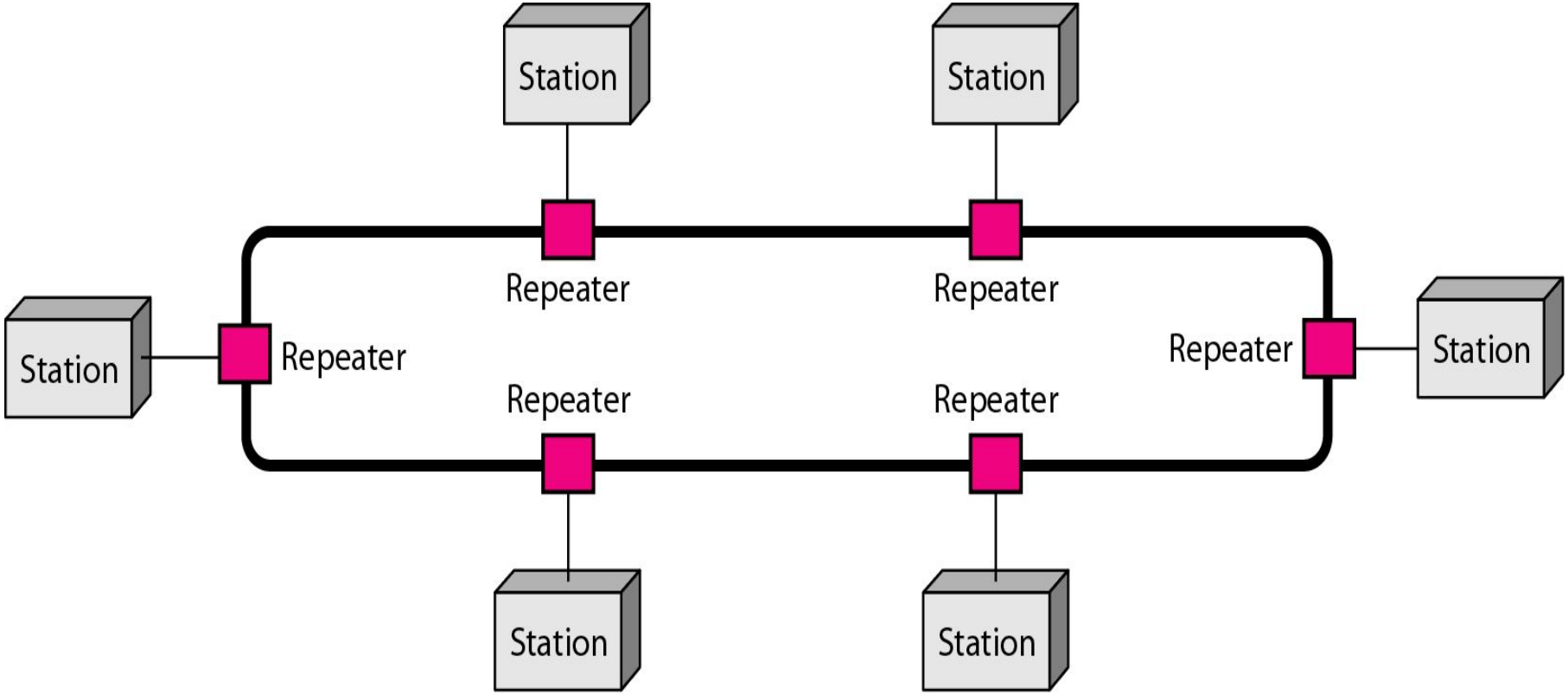
RING TOPOLOGY

It is called ring topology because it forms a ring as each computer is connected to another computer, with the last one connected to the first. Exactly two neighbors for each device.

Features of Ring Topology

- The transmission is unidirectional
- Data is transferred in a sequential manner that is bit by bit. Data transmitted, has to pass through each node of the network, till the destination node.

Ring Topology



Advantages of Ring Topology

- Transmitting network is not affected by high traffic or by adding more nodes, as only the nodes having tokens can transmit data.
- Cheap to install and expand

Disadvantages of Ring Topology

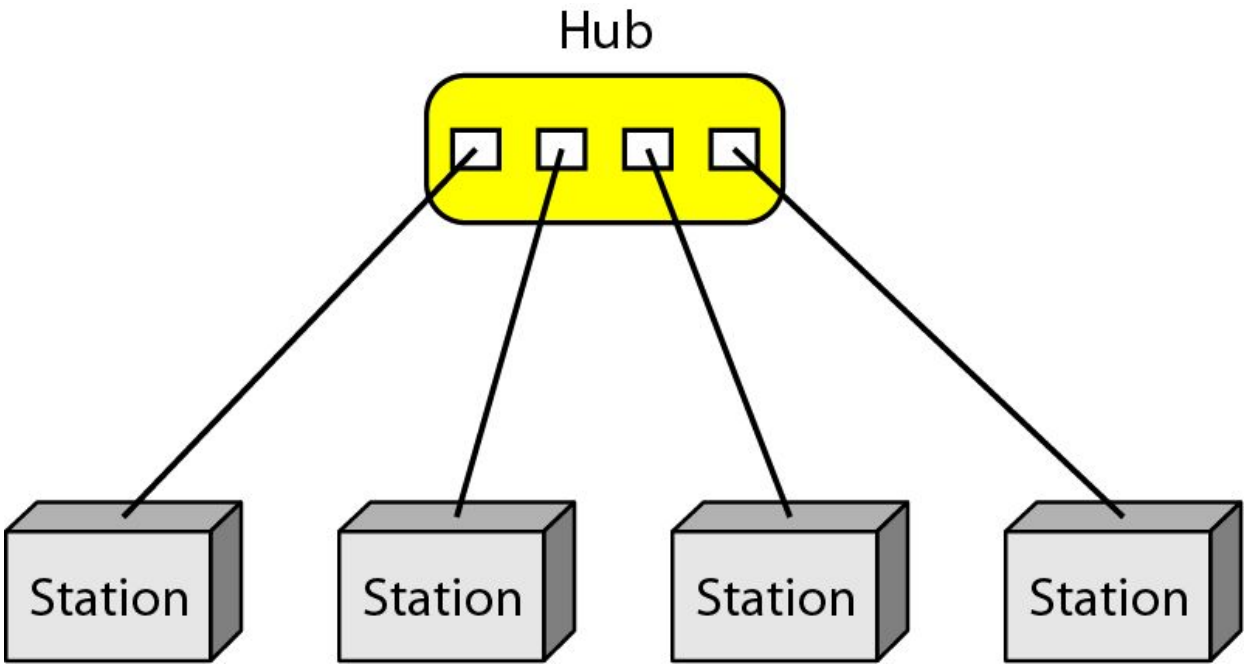
- Troubleshooting is difficult in ring topology.
- Adding or deleting the computers disturbs the network activity.
- Failure of one computer disturbs the whole network.

STAR TOPOLOGY

In this type of topology all the computers are connected to a single hub through a cable. This hub is the central node and all others nodes are connected to the central node.

Features of Star Topology

- Every node has its own dedicated connection to the hub.
- Hub acts as a repeater for data flow.
- Can be used with twisted pair, Optical Fiber or coaxial cable.



Star Topology

Advantages of Star Topology

- Fast performance with few nodes and low network traffic.
- Hub can be upgraded easily.
- Easy to troubleshoot.
- Easy to setup and modify.
- Only that node is affected which has failed, rest of the nodes can work smoothly.

Disadvantages of Star Topology

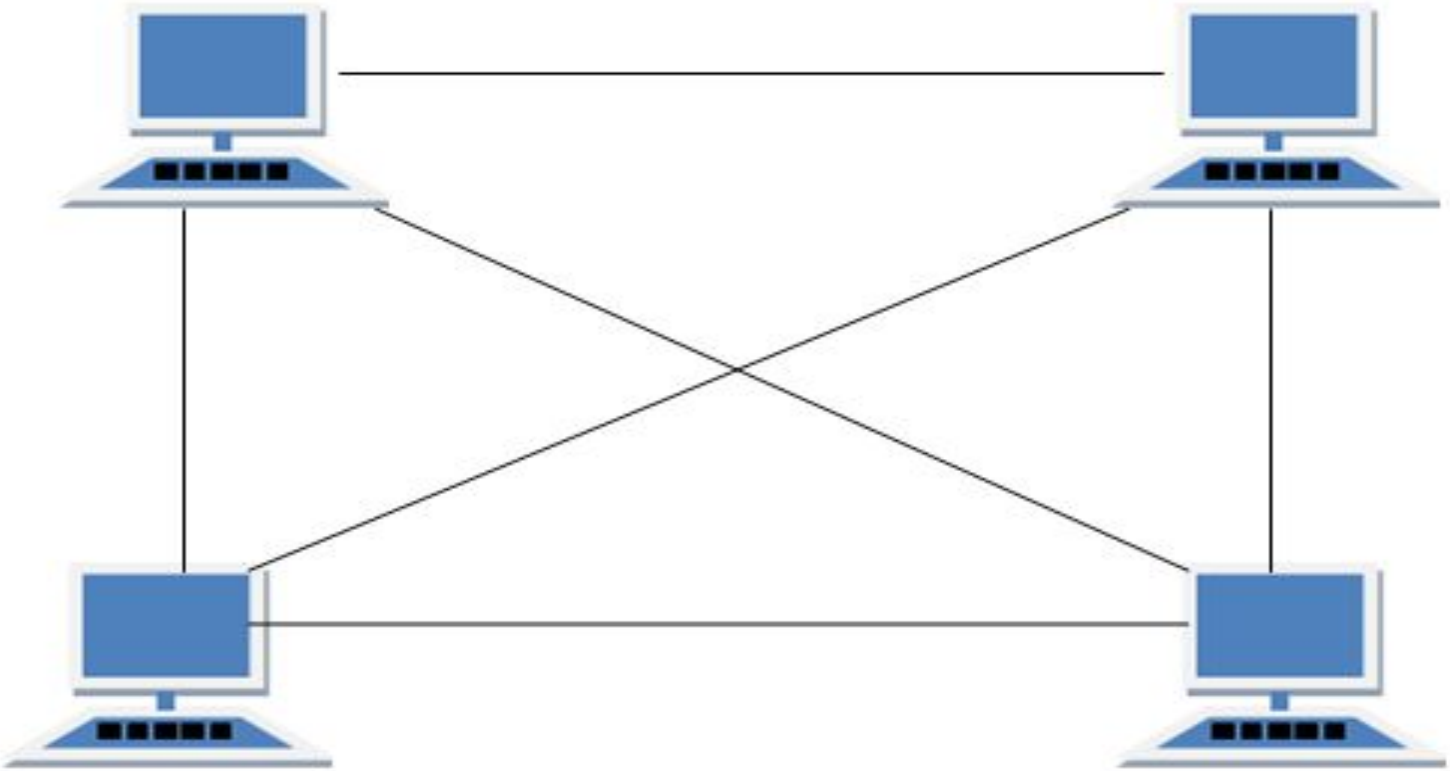
- Cost of installation is high.
- Expensive to use.
- If the hub fails then the whole network is stopped because all the nodes depend on the hub.
- Performance is based on the hub that is it depends on its capacity

MESH TOPOLOGY

It is a point-to-point connection to other nodes or devices. All the network nodes are connected to each other. Mesh has n(n-1)/2 physical channels to link n devices.

Features of Mesh Topology

- Fully connected
- Robust.
- Not flexible.



Mesh Topology

Advantages of Mesh Topology

- Each connection can carry its own data load.
- It is robust.
- Fault is diagnosed easily.
- Provides security and privacy.

Disadvantages of Mesh Topology

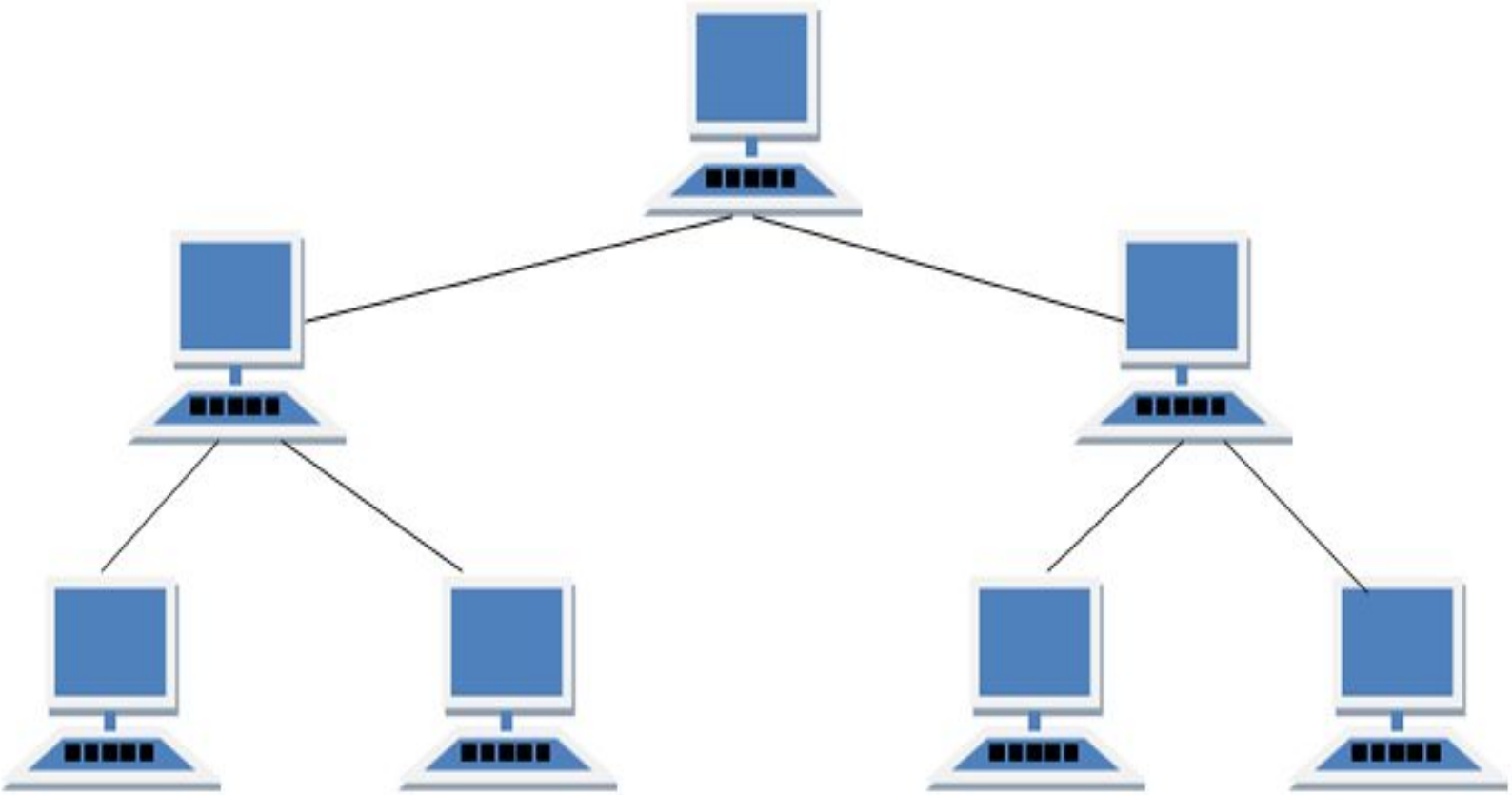
- Installation and configuration is difficult.
- Cabling cost is more.
- Bulk wiring is required.

TREE TOPOLOGY

It has a root node and all other nodes are connected to it forming a hierarchy. It is also called hierarchical topology. It should at least have three levels to the hierarchy.

Features of Tree Topology

- Used in Wide Area Network.
- Expansion of nodes is possible and easy.



Tree Topology

Advantages of Tree Topology

- Extension of bus and star topologies.
- Expansion of nodes is possible and easy.
- Easily managed and maintained.
- Error detection is easily done.

Disadvantages of Tree Topology

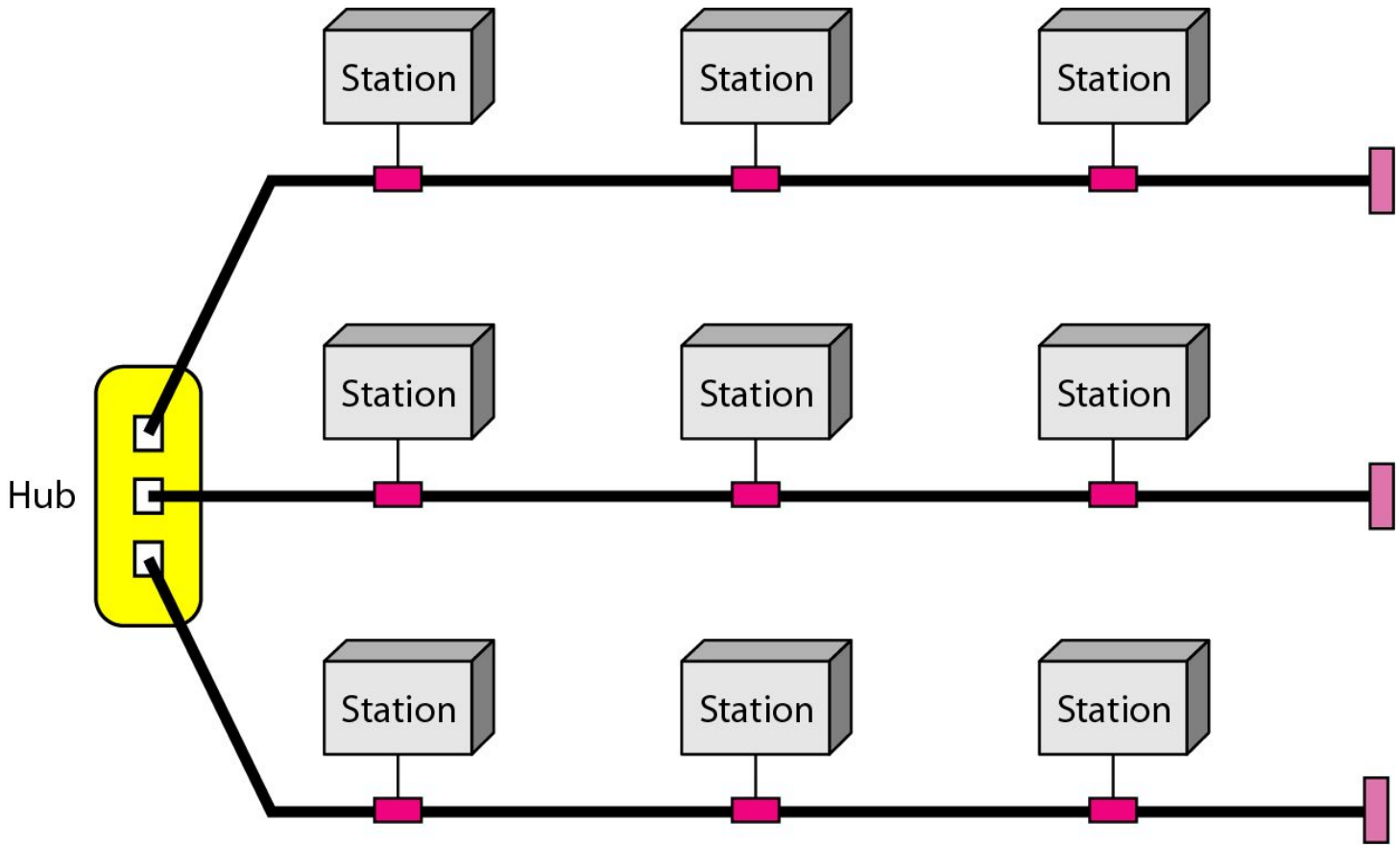
- Heavily cabled.
- Costly.
- If more nodes are added maintenance is difficult.
- Central hub fails, network fails.

HYBRID TOPOLOGY

- It is different type of topology which is a mixture of two or more topologies.
- For example if in an office in one department ring topology is used and in another star topology is used, connecting these topologies will result in Hybrid Topology (ring topology and star topology).

Features of Hybrid Topology

- It is a combination of two or topologies
- Inherits the advantages and disadvantages of the topologies included



Hybrid Topology

Advantages of Hybrid Topology

- Reliable as Error detecting and trouble shooting is easy.
- Effective
- Scalable as size can be increased easily.
- Flexible

Disadvantages of Hybrid Topology

- Complex in design.
- Costly

PROTOCOLS

A protocol is synonymous with rule. It consists of a set of rules that govern data communications. It determines what is communicated, how it is communicated and when it is communicated. The key elements of a protocol are syntax, semantics and timing.

ELEMENTS OF A PROTOCOL

- **Syntax**
  - Structure or format of the data
  - Indicates how to read the bits - field delineation
- **Semantics**
  - Interprets the meaning of the bits
  - Knows which fields define what action
- **Timing**
  - When data should be sent and what
  - Speed at which data should be sent or speed at which it is being received.

NETWORK MODEL

LAYERED TASKS

- Computer networks are complex systems
  - Tasks involve varieties of hardware and software components, and protocols
- Networking task is divided into several subtasks, or layers.

WHY LAYERING?

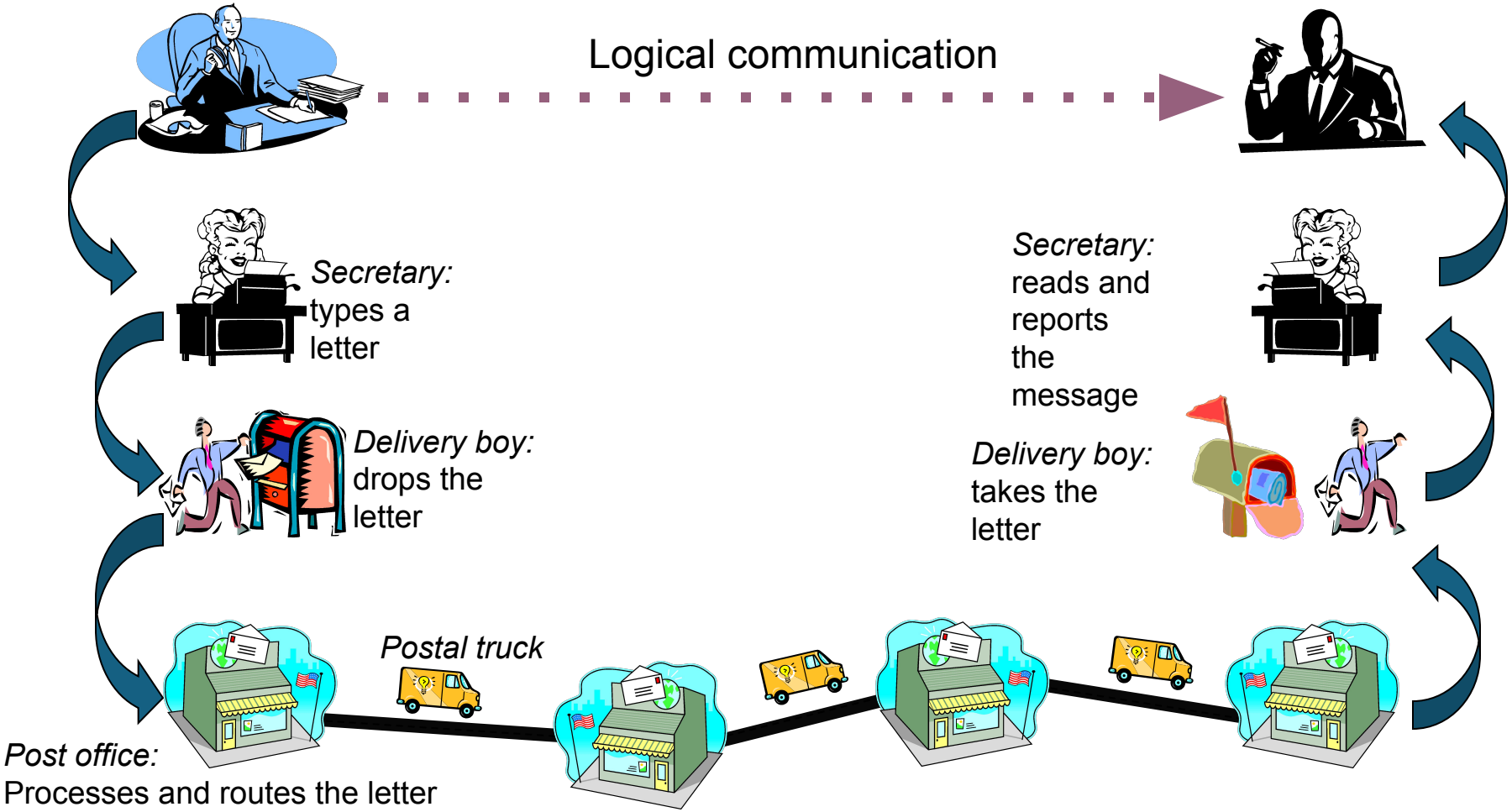
- Networks are complex!
- We need a way to organize the structure of network functionalities and **to reduce the design complexities**
- Benefits of layering
  - ✔ Interoperability
  - ✔ Reuse
  - ✔ Hiding underlying details

REAL WORLD EXAMPLE

- **Communication between managers of two companies**

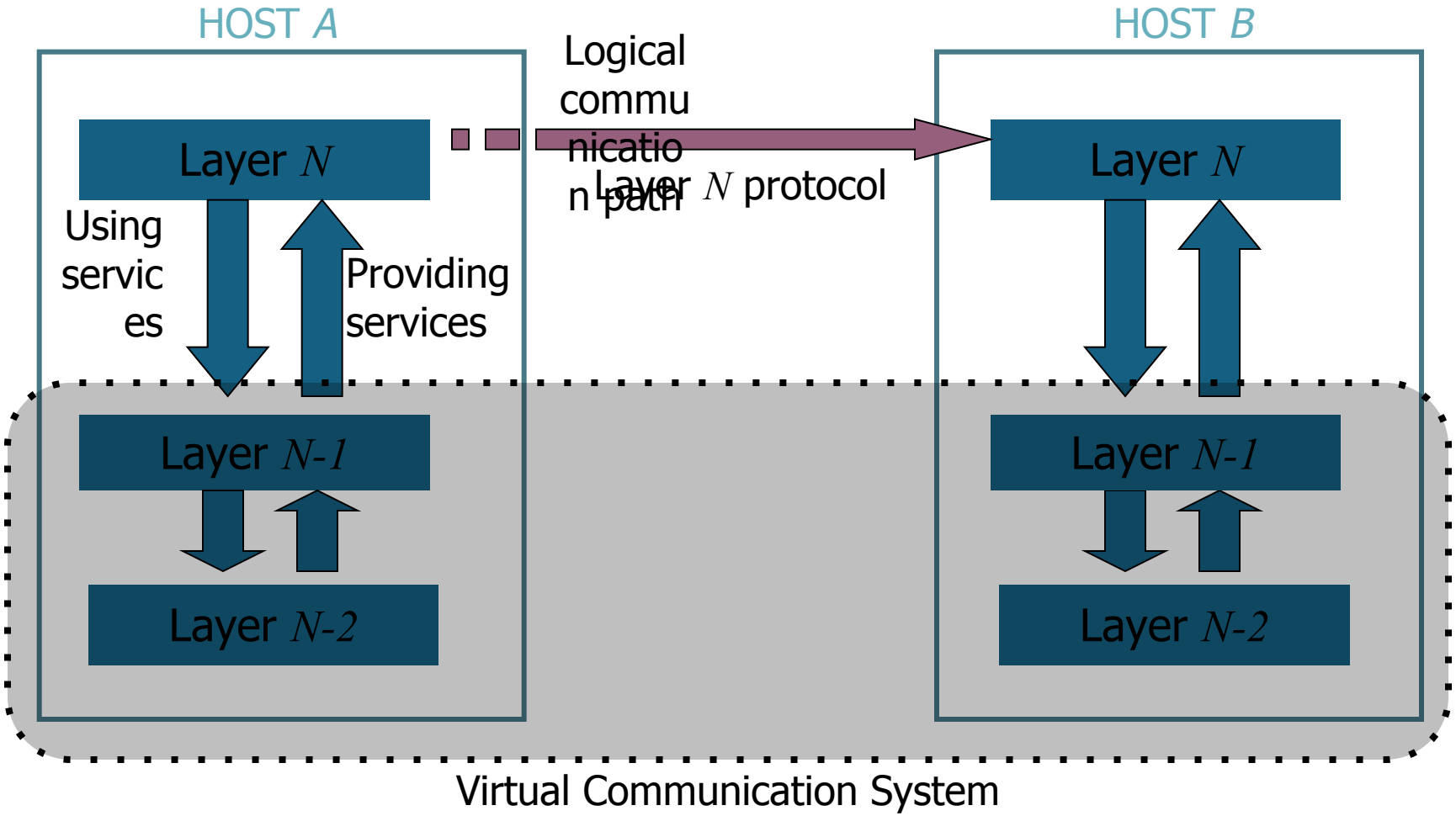
LOGICAL CONNECTION

- Communication takes place through many layers



LAYER MODEL

- Layer *N* uses services provided by Layer *N*-1



NETWORK MODEL

- Network models are responsible for establishing a connection among the sender and receiver and transmitting the data.

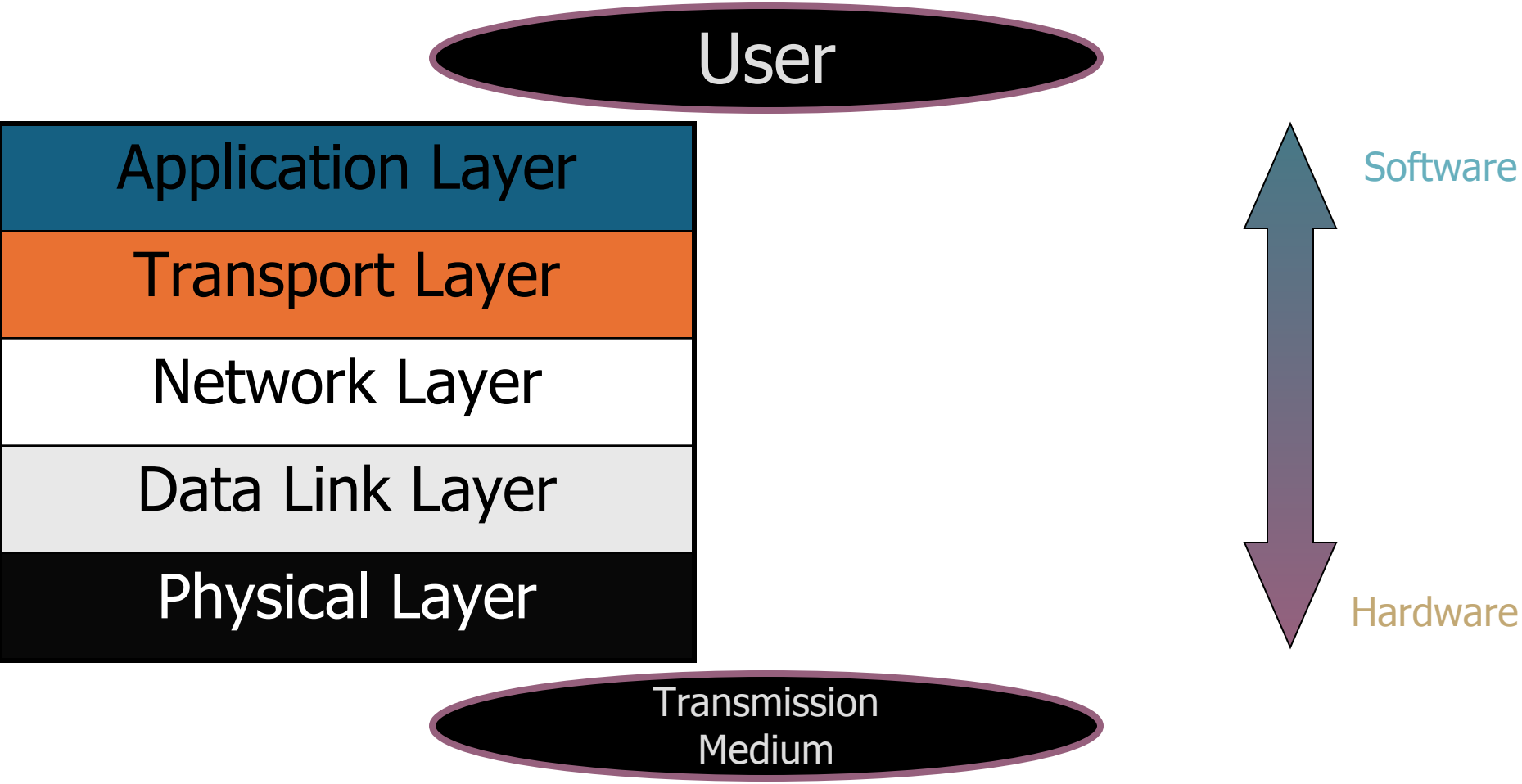
- There are two computer network models on which data communication process relies :
  - TCP/IP Model
  - ISO/OSI Model

TCP/IP MODEL

- The TCP/IP model was developed prior to the OSI model.
- The TCP/IP model is not exactly similar to the OSI model.
- The TCP/IP model consists of five layers: the application layer, transport layer, network layer, data link layer and physical layer.
- The first four layers provide physical standards, network interface, internetworking, and transport functions that correspond to the first four layers of the OSI model .
- TCP/IP is a hierarchical protocol made up of interactive modules, and each of them provides specific functionality.

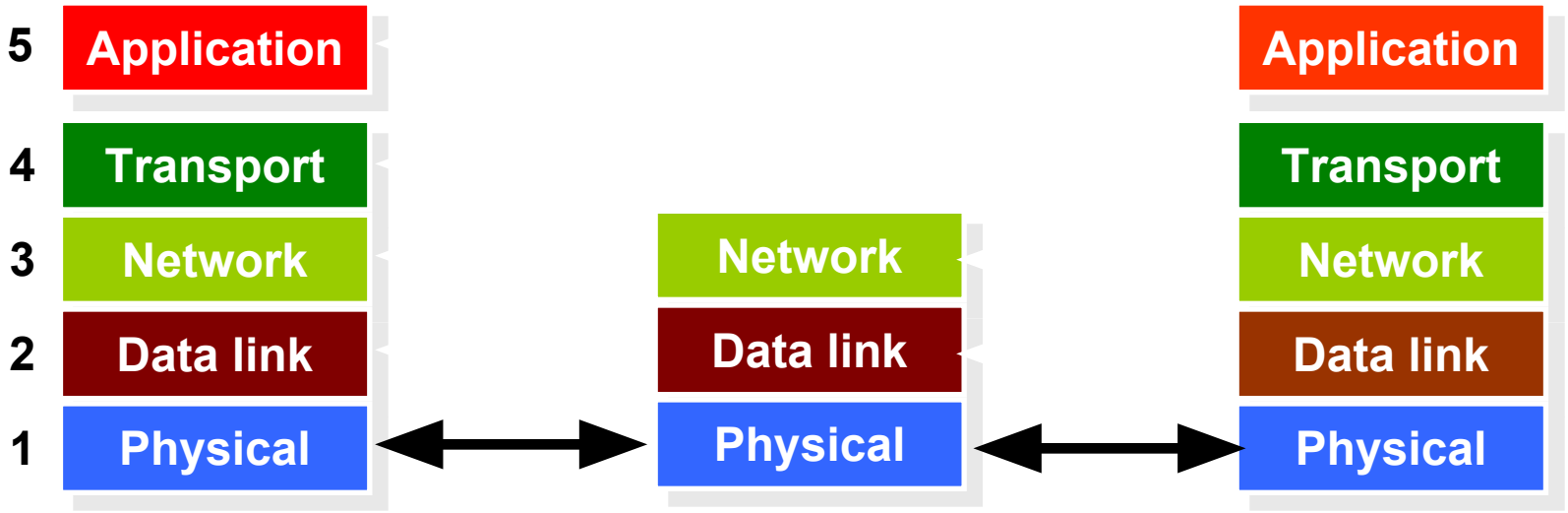
TCP/IP OR INTERNET LAYER MODEL

- **The Internet Protocol Stack**



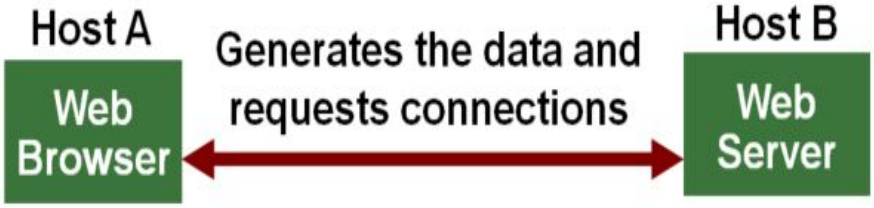
TCP/IP Model: the Internet model

- Each layer relies on services from layer below
- Each layer exports services to layer above
- Interface between layers defines interaction
  - Hides implementation details
  - Layers can change without disturbing other layers



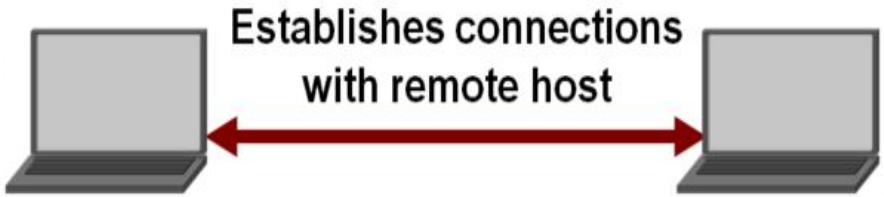
5 Application Layer

The Application layer is the group of applications requiring network communications.



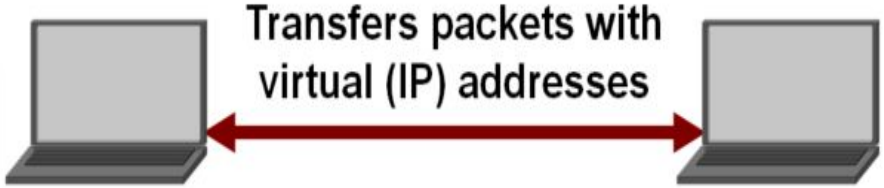
4 Transport Layer (TCP/UDP)

The Transport layer establishes the connection between applications on different hosts.



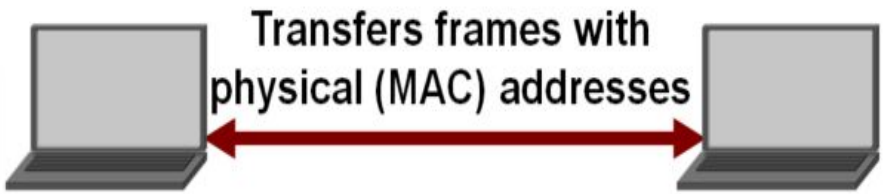
3 Network Layer (IP)

The Network layer is responsible for creating the packets that move across the network.



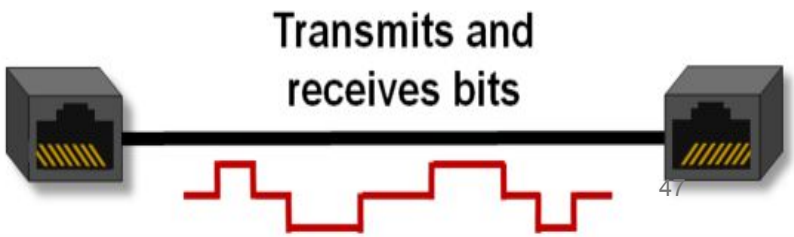
2 Data Link Layer (MAC)

The Data Link layer is responsible for creating the frames that move across the network.



1 Physical Layer

The Physical layer is the transceiver that drives the signals on the network.

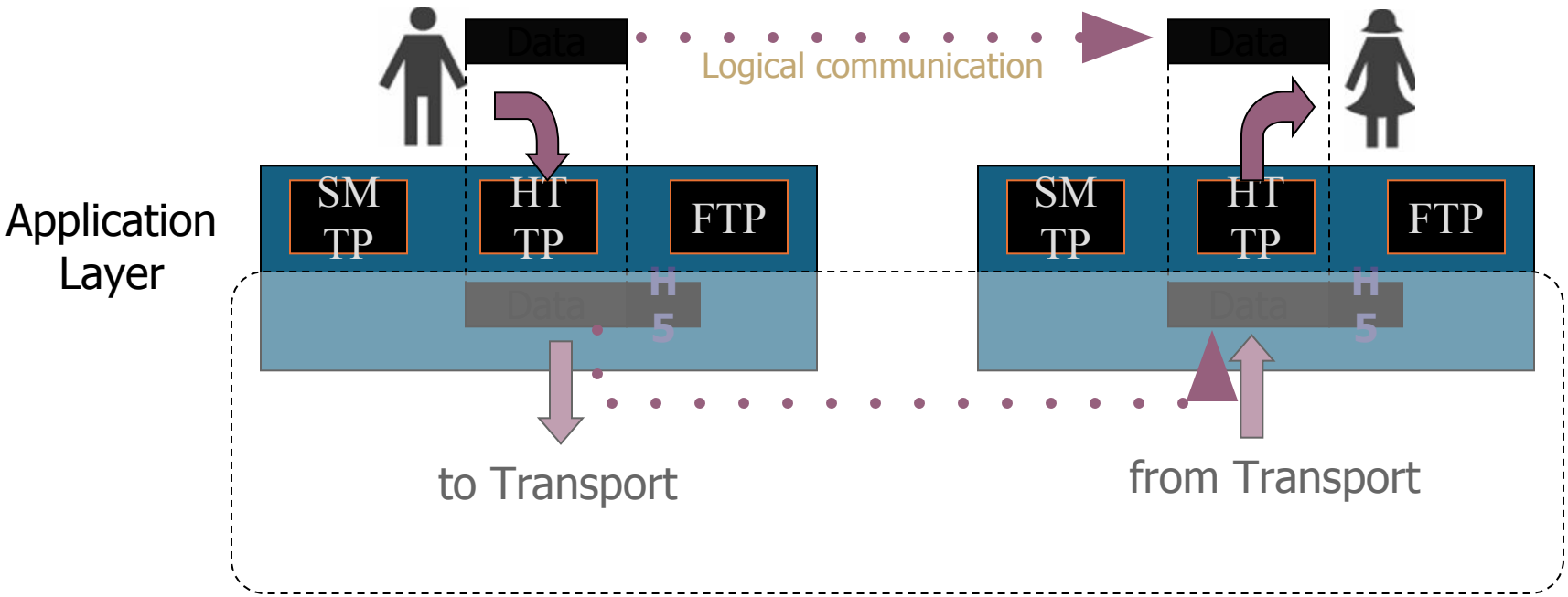


APPLICATION LAYER

- It is the topmost layer in the TCP/IP model.
- It is responsible for handling high-level protocols, issues of representation.
- This layer allows the user to interact with the application.
- It facilitates the user to use the services of the network.
- It is used to develop network-based applications.
- It provides user services like user login, naming network devices, formatting messages, and e-mails, transfer of files etc.
- When one application layer protocol wants to communicate with another application layer, it forwards its data to the transport layer.

APPLICATION LAYER

- The only layer to interact with user



Responsible for providing services to the user

TRANSPORT LAYER

- The transport layer is a 4th layer from the top.
- It is responsible for end to end data transmission service using connection oriented or either through connection-less protocols.
- TCP and UDP are two transport layer protocols that provide a different set of services to the network layer.
- It provide multiplexing/demultiplexing service.
- It also provides other services such as reliable data transfer, bandwidth guarantees, and delay guarantees,flow control.

TRANSPORT LAYER

Responsible for delivery of a message from one process to another

- Duties/services
  - Service-point addressing/Port Address
  - Segmentation and reassembly
  - Connection control
  - Flow control (end-to-end)
  - Error control (end-to-end)

Transport Layer Services

- **Segmentation** : **Large data is divided into smaller segments** at the senders end and then these smaller segments are recombined in exact format before it is received at receivers end.

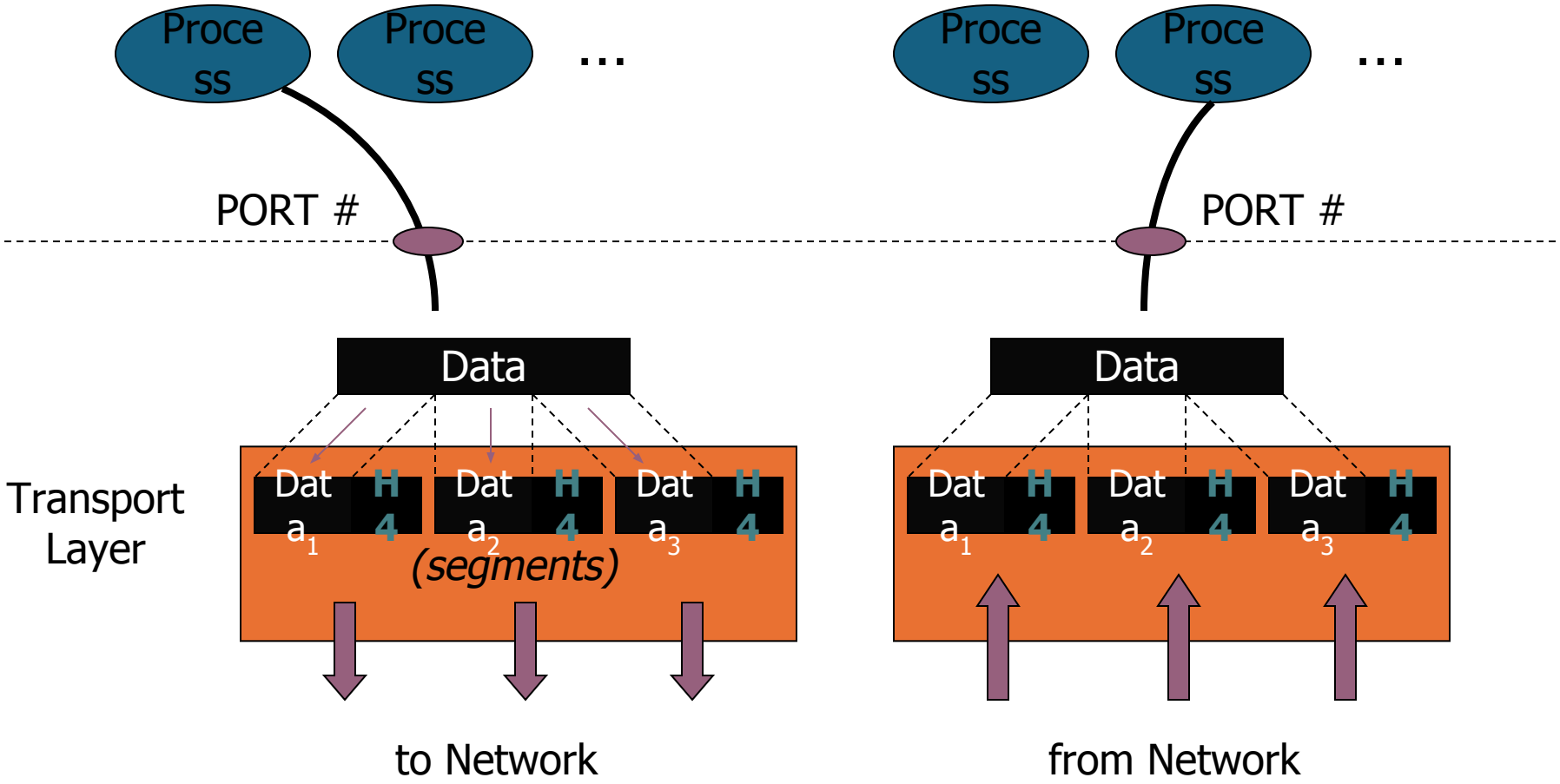
- **Error Control** : Distortion, Un-delivery of data packets and noise in data packets results in error and this is prevented using Data Link and this feature is called Error Control.

- **Flow Control** : Transport-Layer ensures that faster data transmission and slow data absorption or slow data transmission and faster data absorption must not occur.

- Transport-Layer performs Multiplexing & De-multiplexing.

- **Connection Control**: Connection oriented service using **TCP**(Transmission Control Protocol) Protocol & Connection less service using **UDP**(User Data-gram Protocol) Protocol are performed by transport layer.

Transport Layer

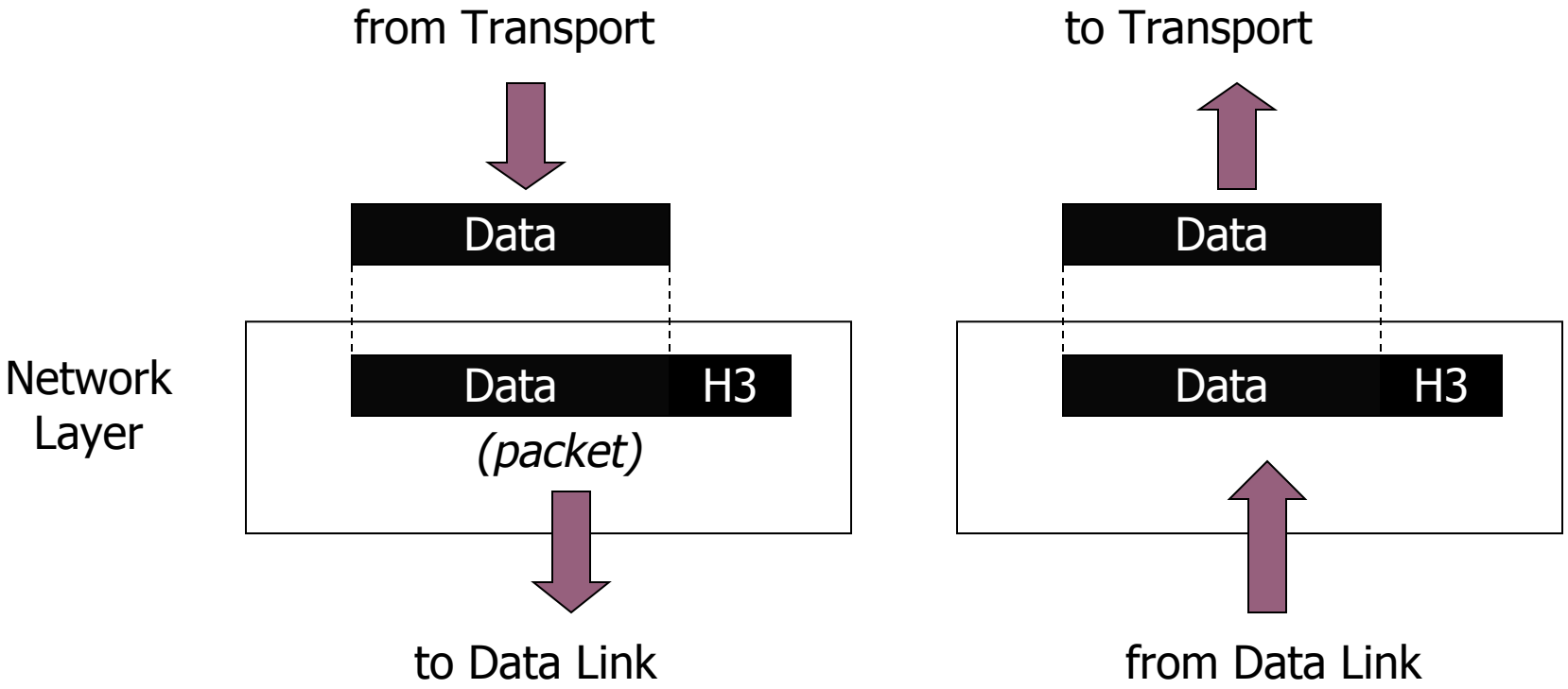


NETWORK LAYER

Responsible for the delivery of packets from the original source to the destination

- Duties/services
  - Logical addressing
  - Routing

NETWORK LAYER



DATA LINK LAYER

- ✔ Framing: The data link layer divides the stream of bits received from the network layer into manageable data units called frames.

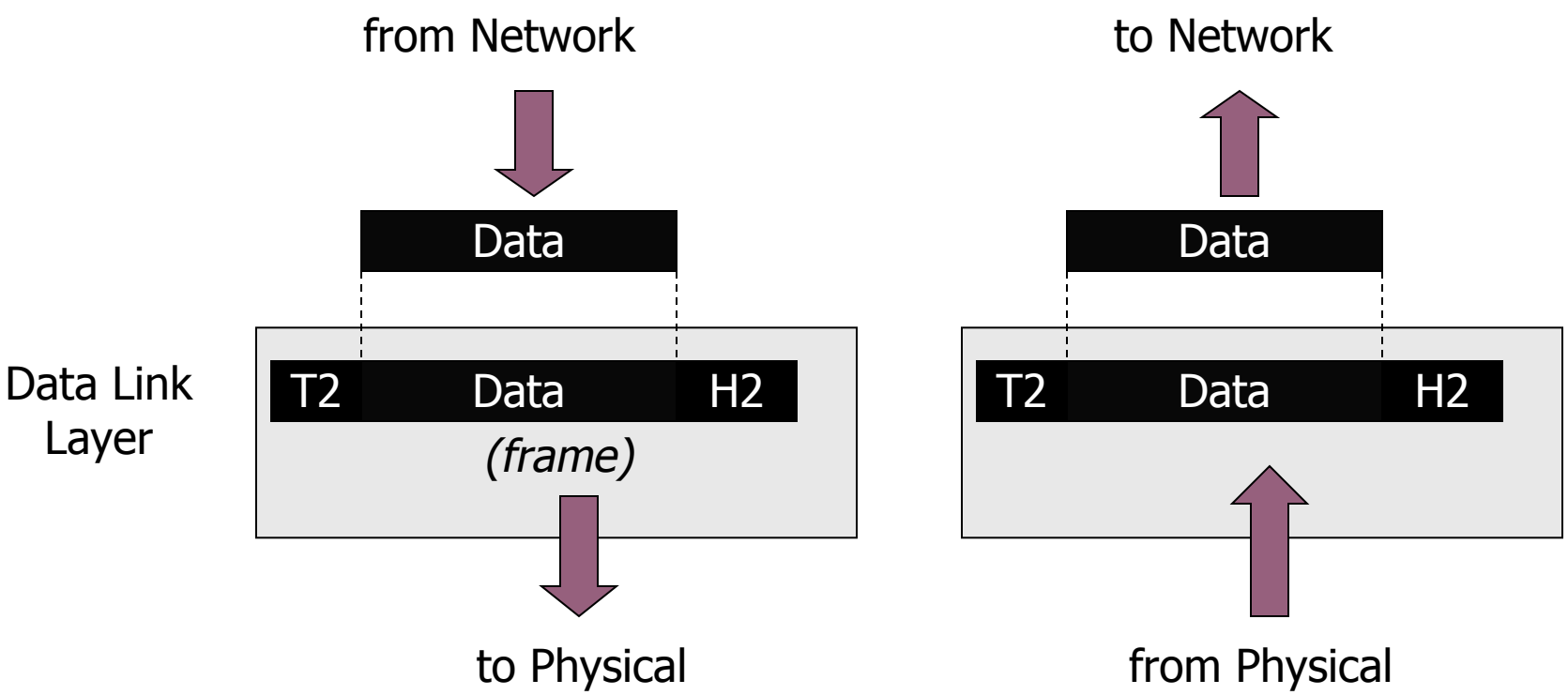
- ✔ Physical addressing: If frames are to be distributed to different systems on the network, the data link layer adds a header to the frame to define the sender and/or receiver of the frame. If the frame is intended for a system outside the sender's network, the receiver address is the address of the device that connects the network to the next one.

- ✔ Flow control (hop-to-hop): A receiving node can receive the frames at a faster rate than it can process the frame. Without flow control, the receiver's buffer can overflow, and frames can get lost. To overcome this problem, the data link layer uses the flow control to prevent the sending node on one side of the link from overwhelming the receiving node on another side of the link.

- ✔ Error control (hop-to-hop):The data link layer adds reliability to the physical layer by adding mechanisms to detect and retransmit damaged or lost frames. It also uses a mechanism to recognize duplicate frames. Error control is normally achieved through a trailer added to the end of the frame.

- ✔ Access control:. When two or more devices are connected to the same link, data link layer protocols are necessary to determine which device has control over the link at any given time.

DATA LINK LAYER



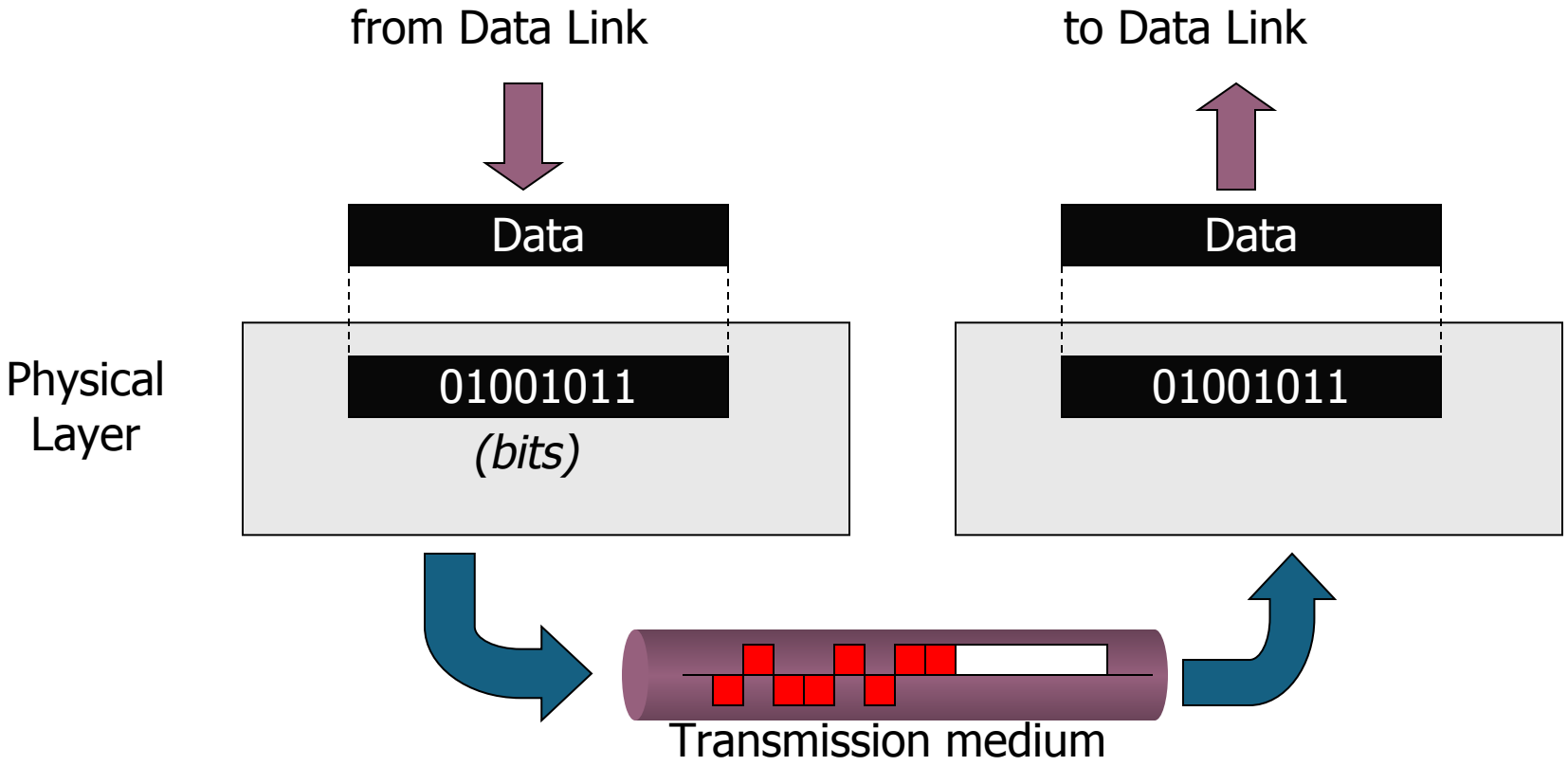
Responsible for transmitting frames from one hop (node) to the next

PHYSICAL LAYER

The physical layer is responsible for movements of individual bits from one hop (node) to the next

- Physical Layer is the bottom-most layer.
- It is associated with the electrical, mechanical and transmission media .
- Physical layer deals with all the physical devices that can be used for data communication.

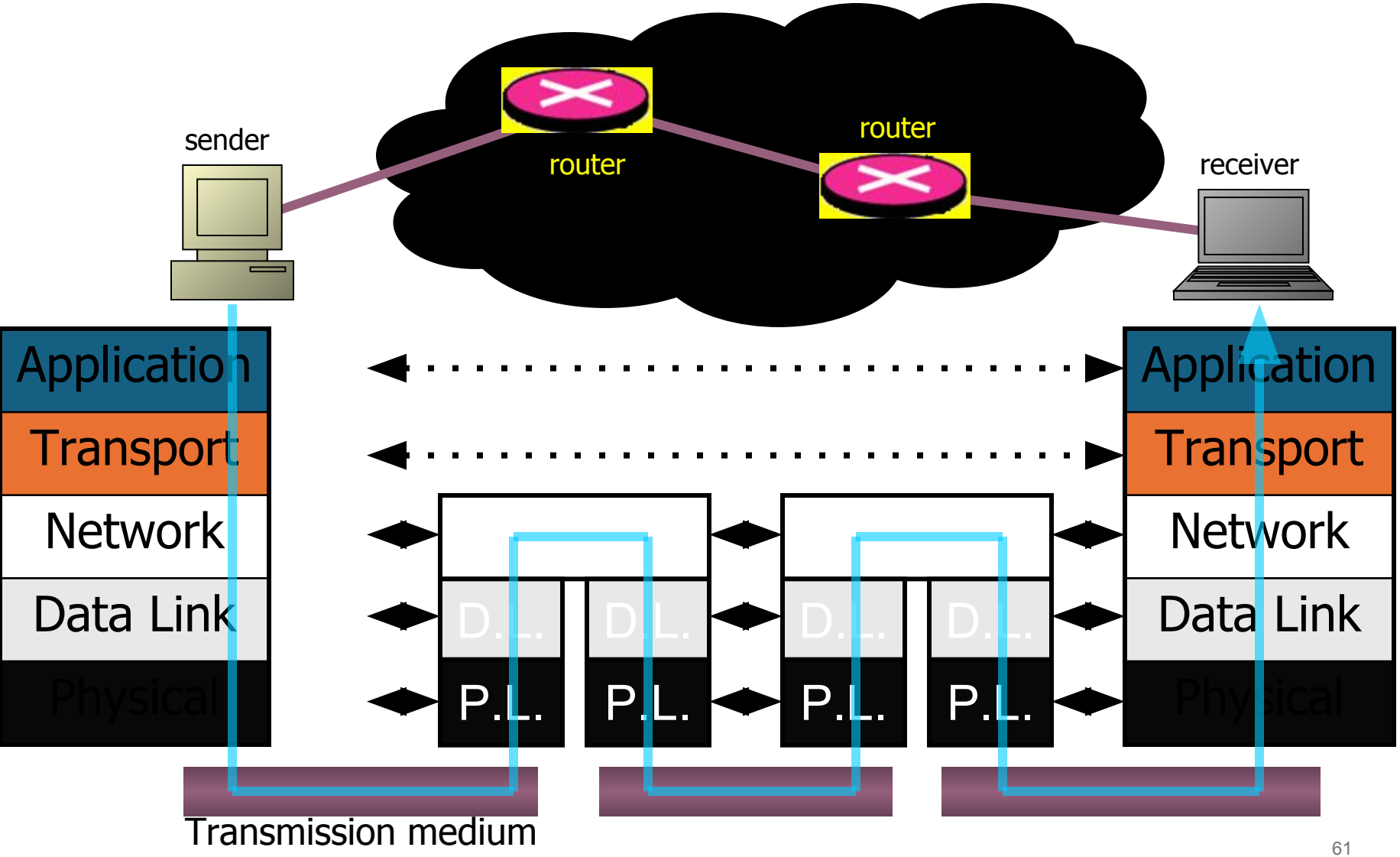
PHYSICAL LAYER



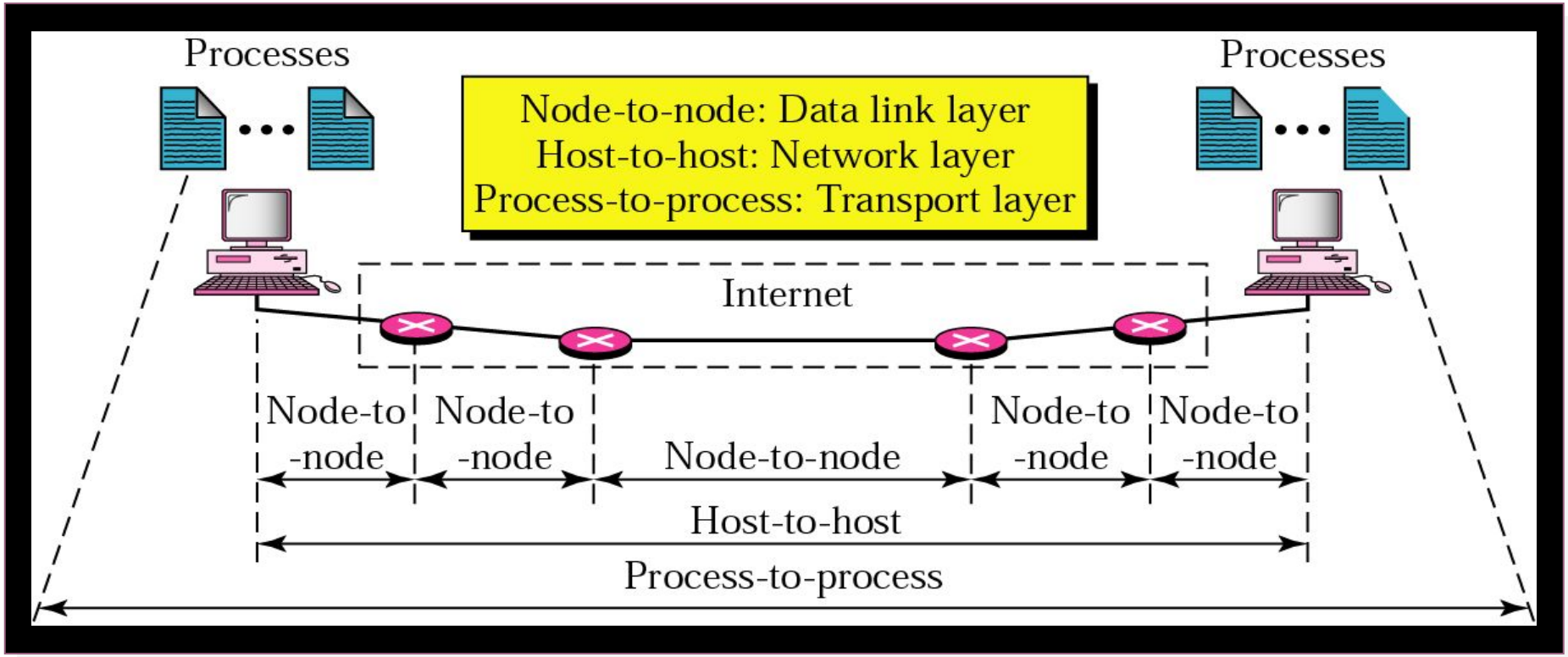
Physical Layer Functions

- Physical characteristics of interfaces and medium
- Representation of bits: Data consists of a stream of bits (sequence of 0s or 1s) with no interpretation.
- Data rate: The number of bits sent per second.
- Synchronization of bits: the sender and the receiver clocks must be synchronized.
- Line configuration: Point to Point or Multipoint
- Physical topology: Bus/ Star/Mesh…
- Transmission mode: Simplex/Half/Full

INTERNET MODEL



INTERNET MODEL



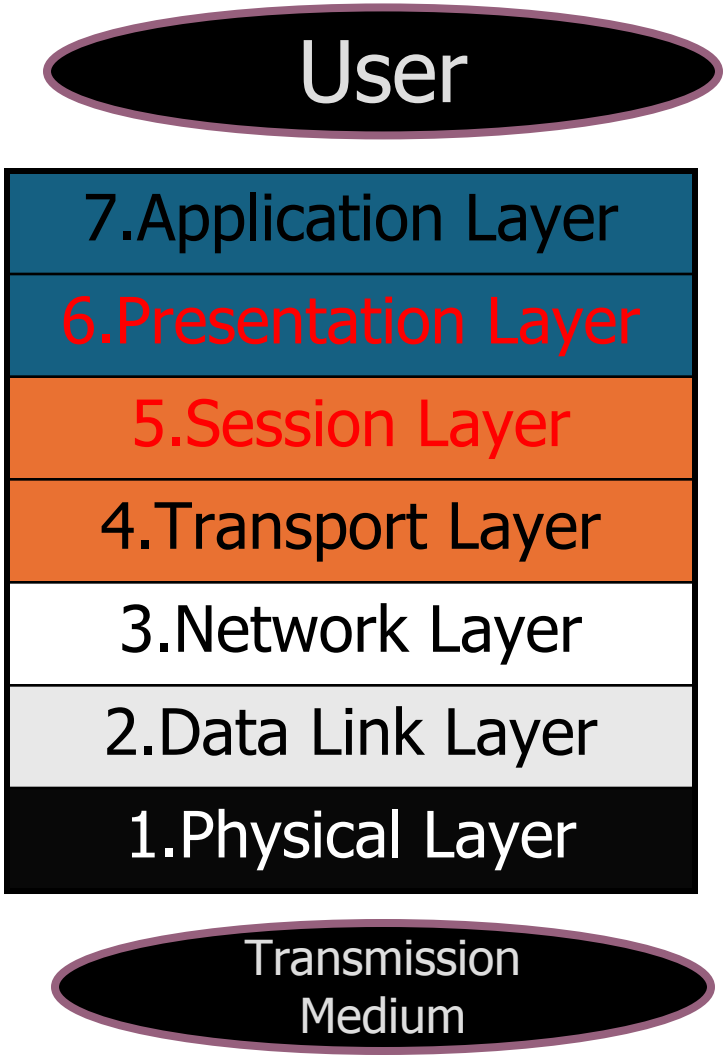
INTERNET PROTOCOL SUITE

Layer	Protocols
Application	HTTP, FTP, Telnet, SMTP, ...
Transport	TCP, UDP, SCTP, ...
Network	IP (IPv4), IPv6, ICMP, IGMP, ...
Data Link	Ethernet, Wi-Fi, PPP, ...
Physical	RS-232, DSL, 10Base-T, ...

OSI MODEL

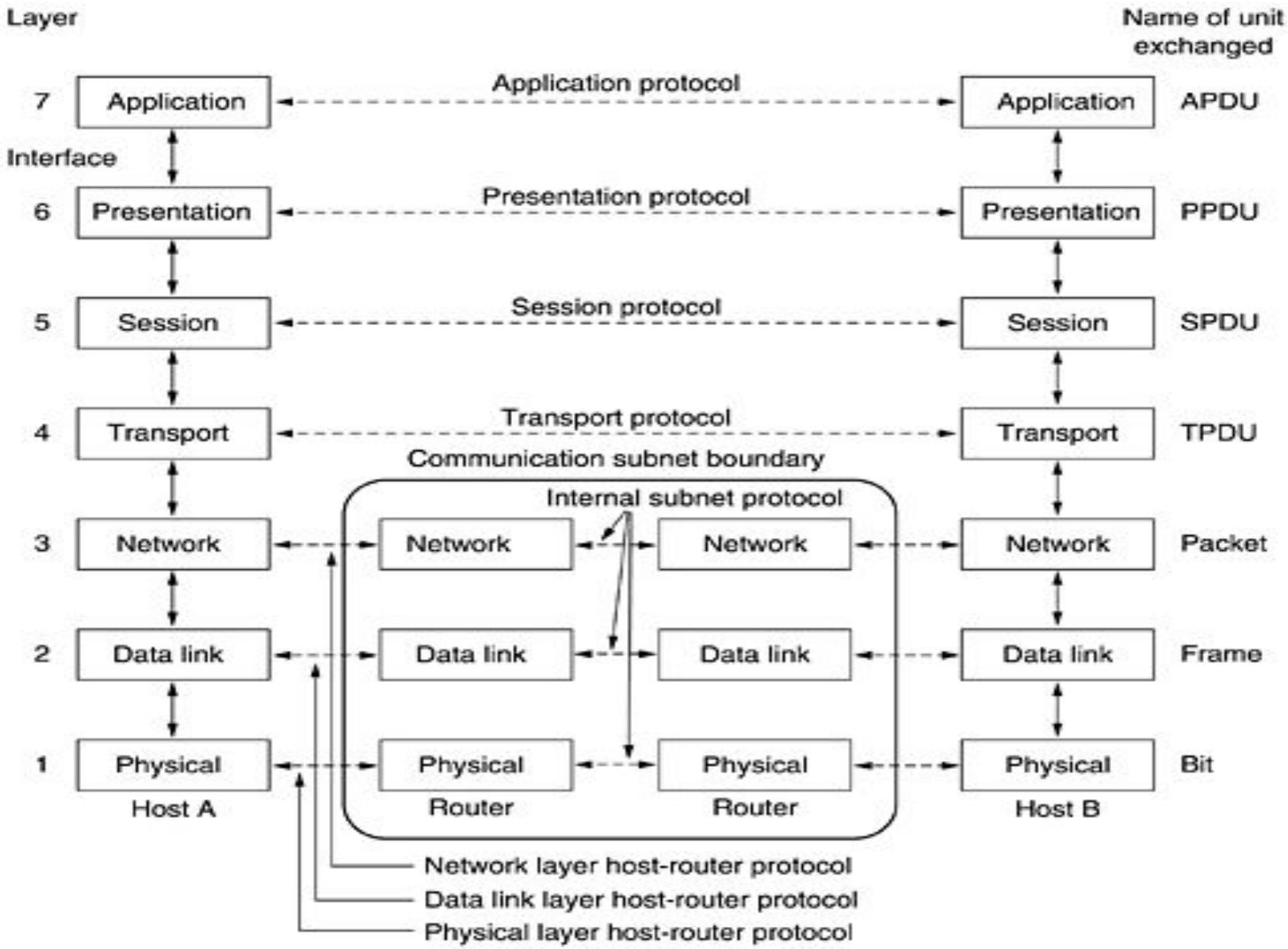
- OSI stands for Open System Interconnection.
- Before networks came into existence, computers could only communicate from the same manufacturers.
- OSI was designed by ISO (International Organization for Standardization) in late 1970's.
- To break this barrier so that Computers from different manufacturer could communicate with each other.
- OSI is a "Layered Approach", which is a logical representation of how Data Communication should occur.

OSI Model

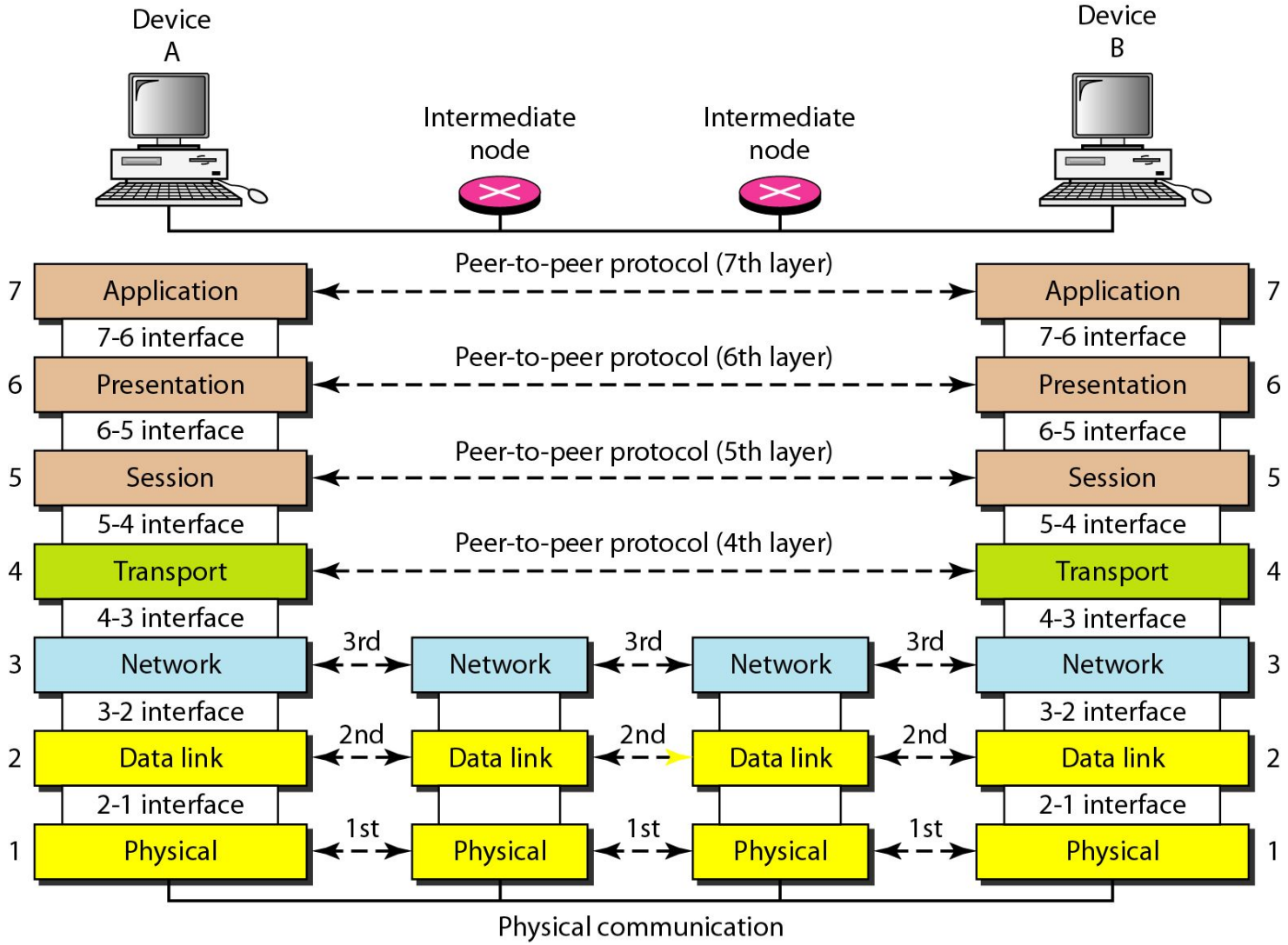


- OSI has 7 different layers
- Two new layers is added
  - Presentation Layer
  - Session Layer
- It is divided into two groups,
  - Top 3 layers defines how application within end stations will communicate with each other and with the user (Software Layer).
  - Bottom 4 layers defines how actually data is transferred (Hardware Layer).

The OSI Reference Model



The interaction between layers in the OSI model



SESSION LAYER

- This layer is responsible for establishment of connection, maintenance of sessions, authentication and also ensures security.

- Duties/services
  - **Dialog Controller** :
    - Allows the two processes to establish, use and terminate a connection.
    - Communication in half-duplex or full-duplex.
  - **Synchronization :**
    - allows a process to add checkpoints
    - data is re-synchronized
    - Session recovery

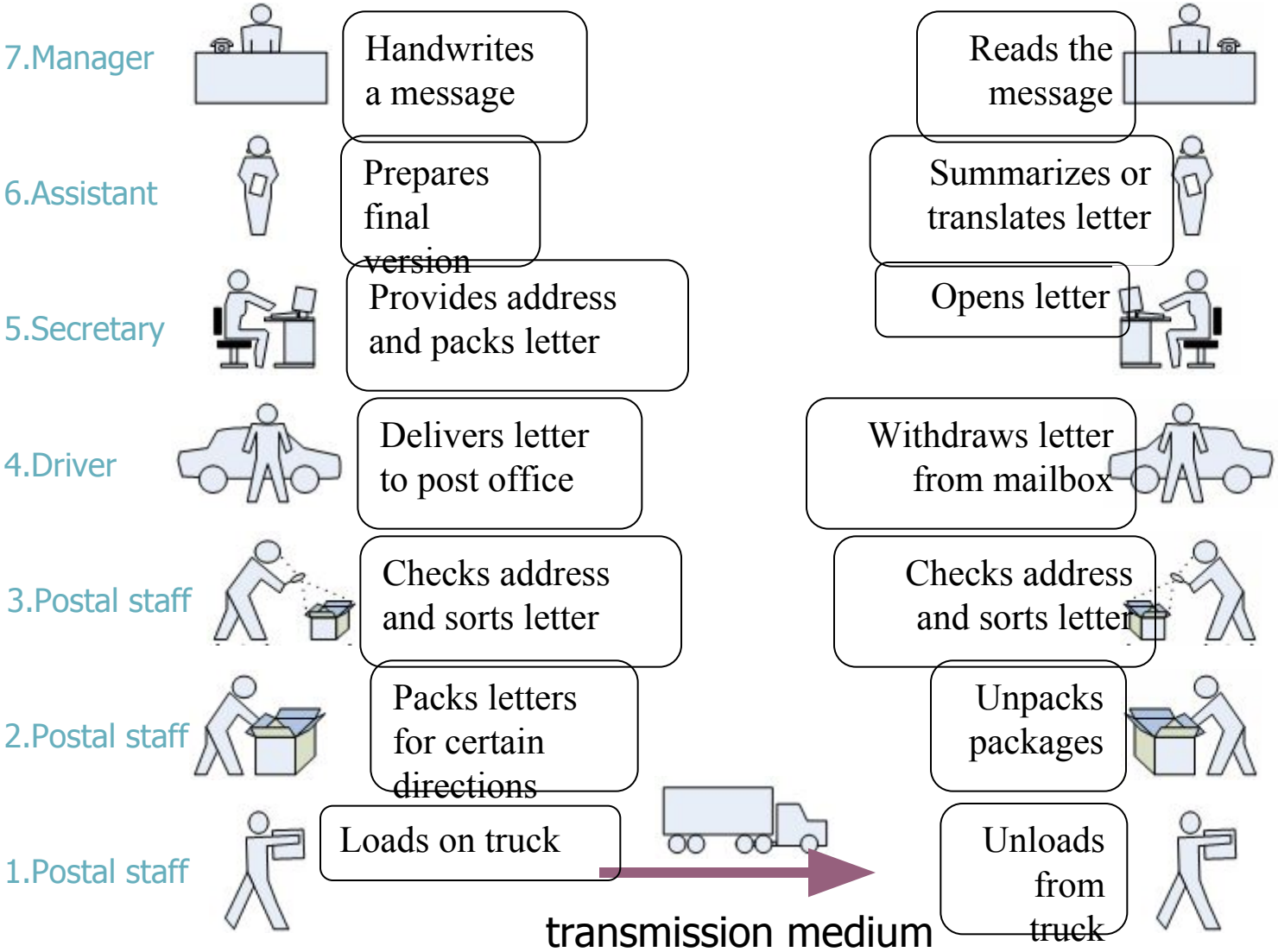
PRESENTATION LAYER

- A Presentation layer is mainly concerned with the syntax and semantics of the information exchanged between the two systems.
- It acts as a data translator for a network.
- This layer is a part of the operating system that converts the data from one presentation format to another format.
- The Presentation layer is also known as the syntax layer.

PRESENTATION LAYER FUNCTIONS

- Duties/services
  - Data translation: For example, ASCII to EBCDIC.
  - Encryption/Decryption
  - Compression: Reduces the number of bits that need to be transmitted on the network.

OSI LAYERS IN REAL WORLD



Pictures from *Wikipedia*

OSI	TCP/IP
It is developed by ISO (International Standard Organization)	It is developed by ARPANET (Advanced Research Project Agency Network).
OSI refers to Open Systems Interconnection.	TCP refers to Transmission Control Protocol.
Session and presentation layers are a part of the OSI model.	There is no session and presentation layer in the TCP model.
OSI is less reliable	TCP/IP is more reliable

END OF MODULE 1.1